

Beamforming for Ultra Reliable Low Latency Wireless Communication 5G/6G Systems

Sagar S. Jagtap

All India Shri Shivaji Memorial Society's, Institute of
Information Technology
Pune, Maharashtra 411001 India
sagar.jagtap@aissmsioit.org

Manas M. Pandit

All India Shri Shivaji Memorial Society's, Institute of
Information Technology
Pune, Maharashtra 411001 India
panditmanas390@gmail.com

Shrikrishna Kolhar

Symbiosis Institute of Technology, Pune Campus
Symbiosis International (Deemed University)
Pune, Maharashtra 412115, India
shrikrishna.kolhar@sitpune.edu.in

Sujal V. Pardakhe

All India Shri Shivaji Memorial Society's, Institute of
Information Technology
Pune, Maharashtra 411001 India
sujalpardakheytl@gmail.com

Abstract—Ultra-Reliable Low-Latency Communication (URLLC) is a key enabling technology for critical 5G/6G mission-critical applications such as autonomous vehicles and remote healthcare. To achieve its demanding sub-millisecond latency and 10^{-5} packet error rate targets, highly optimized physical layer solutions are necessary. Beamforming specifications are able to meet these requirements by focusing signal energy to provide improved link reliability, link power, and spectral efficiency. This paper provides a comparative analysis of analog, digital, and hybrid beamforming solutions for URLLC, from a perspective of complexity, speed, and reliability. It discusses machine learning frameworks for dynamic channel conditions and user mobility. Finally, we discuss remaining challenges such as hardware constraints and massive MIMO/mmWave integration, with the assertion that sophisticated beamforming will be critical for maximizing performance of future wireless networks.

Index Terms—Beamforming, URLLC, 5G, 6G, Machine Learning, Massive MIMO, AAR-Net, Low Latency, Wireless Communication.

I. INTRODUCTION

The development of mobile networks has led to more sophisticated categories of services, such as Ultra-Reliable Low-Latency Communication (URLLC). URLLC is an integral component of mission-critical tasks in 5G and upcoming 6G networks [2]–[4]. It is designed for extremely high levels of performance. URLLC implies a guaranteed end-to-end latency of 1 millisecond or less, along with exceptional reliability that allows losing no more than 10^{-5} to 10^{-7} packets [4], [8]. Such parameters become imperative in applications ranging from self-driving cars to telemedicine, robotic production lines, and power grids. Any latency or data distortion in these applications can have disastrous effects [4], [5].

However, addressing these challenging criteria in an environment characterized by inherent randomness and errors is not an easy task [6], [17], [20]. Ordinary approaches to information transfer do not always provide satisfactory results amid constantly changing circumstances, like user mobility, interferences, and channel dynamics [7].

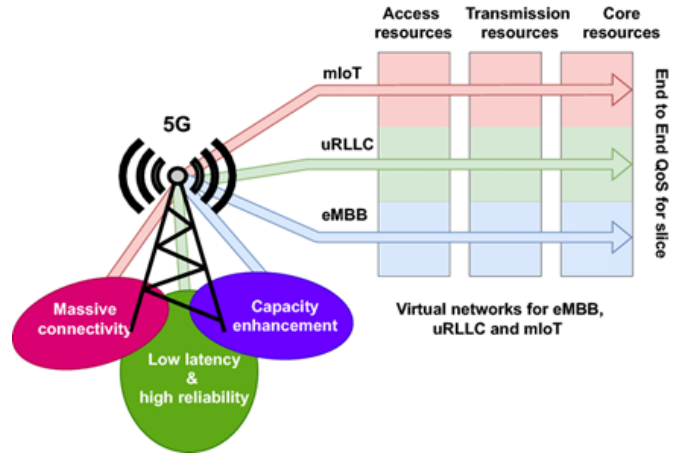


Fig. 1: The Latency-Reliability Plane for 5G/6G Applications.

Beamforming serves as a spatial processing technique that uses antenna arrays to direct the signals' emissions or receptions towards certain target locations. In addition, the use of various analog, digital or hybrid techniques allows one to trade off computational complexity, latency, and power consumption among other issues..

TABLE I: Comparison of 5G/6G Service Classes and Key Performance Indicators (KPIs)

Service Class	E2E Latency	Reliability (PER)	Peak Data Rate	Representative Use Case
<i>eMBB</i>	≥ 4 ms	$\leq 10^{-3}$	High (Gbps)	HD Video, VR/AR
<i>mMTC</i>	High (Flexible)	Variable	Low (kbps/Mbps)	IoT Sensors, Tracking
<i>URLLC</i>	≤ 1 ms	$\leq 10^{-5}$	Med (Mbps/Gbps)	Remote Surgery, Robotics

This paper examines the utilization of the beamform-

ing techniques for meeting the demanding requirements of URLLC within 5G and future 6G communication systems. Specifically, it presents the state-of-the-art technologies that could be employed as well as the advances made recently regarding adaptive beamforming and the use of machine learning. Ultimately, after analyzing some of the important challenges faced by this architecture, this work proposes some possible directions of research towards building extremely reliable and low-latency communication systems. [1], [6], [15].

II. LITERATURE REVIEW

The emergence of 6G out of URLLC entails the transition from classical mathematical modeling to intelligent and hardware-aware approaches. Literature Review: This section outlines research trends in the following areas:

A) URLLC Parameters and Their Limitations: The technology of URLLC is used as the basis for developing mission-critical applications, such as telemedicine, self-driving cars, and robots. According to the set standards, the acceptable E2E delay cannot be more than ≤ 1 ms, whereas the permissible PER equals 10^{-5} . The realization of the specified reliability entails addressing numerous technical difficulties because any delay in the unstable wireless connection can cause devastating consequences [4], [7], [20], [28].

B) Deep Learning for CSI and Beam Prediction: Modern beamforming technologies tend to move away from static designs based on codebooks and proceed with developing dynamic models based on neural networks. Studies show that neural networks are superior to iterative MMSE algorithms in handling complex channel dynamics. For instance, polar coordinates have enabled AAR-Net to preserve valuable phase information lost when using Cartesian coordinates [9], [13], [27].

C) Adaptive Algorithms Featuring Machine Learning Techniques: In order to tackle the adverse effects of user mobility and fast fading channels, machine learning techniques become an indispensable requirement. Although theoretical accuracy is ensured in terms of the algorithms' operation, there is a need for predictable latency times. Fortunately, modern innovations suggest that neural networks can infer in sub-microsecond intervals (of 1.3×10^{-6} seconds). Thus, they avoid the challenge of channel aging due to excessive iterations [11], [14], [18].

D) Massive MIMO and mmWave Technologies: Finally, the last aspect of beamforming involves the effective incorporation of these algorithms into mmWave/Massive MIMO communication systems. They are essential for ensuring the required bandwidth of 6G; however, such communication systems bring up new problems, for instance, hardware limitations and possible beam misalignment. Therefore, modern research focuses on developing fast alignment techniques to reduce latency [?], [1], [5], [15], [16], [21], [24]–[26], [31].

III. BEAMFORMING FUNDAMENTALS

A. Beamforming Principle and Model

As an important physical layer technique, beamforming makes URLLC possible through using antenna arrays to focus the energy of signals on the desired area so that interference can be minimized while spectrum efficiency can be optimized. Beamforming makes use of constructive and destructive interference through control over phase and amplitude of the signals emitted from the antenna arrays [6], [15].

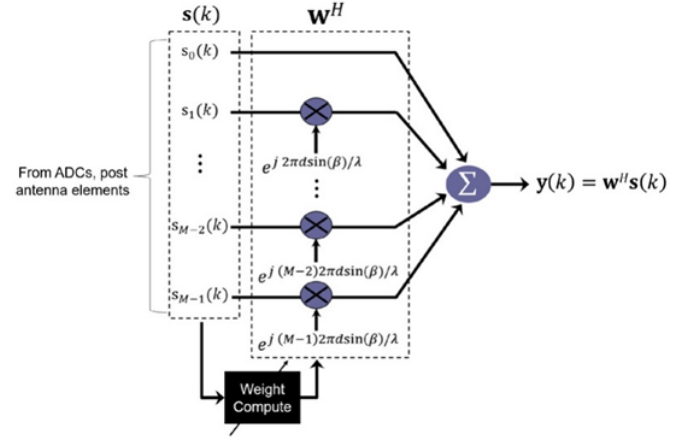


Fig. 2: Beamforming applying weights as sum of products.

It allows for an improved SNR and spatial filtering. Beamforming is essential for 5G/6G communications since it allows for an improved SNR and spatial filtering. When used in the context of URLLC communications, it is particularly useful since it minimizes transmission delay and packet loss so that sub-milliseconds responses can be achieved and maintained [15], [30].

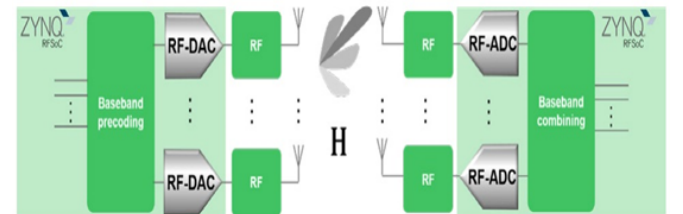


Fig. 3: Channel matrix H for beamforming in wireless communication.

B. URLLC Performance Commitments

Ultra Reliable Low Latency Communication (URLLC) defines several requirements for beamforming techniques. First, E2E delay must not exceed ≤ 1 ms. This means that OTA delay must remain in the range of 0.1 – 0.5 ms, which is necessary for performing real-time activities such as industrial robotics. To ensure service continuity, the PER value cannot exceed $\leq 10^{-5}$. It means that the architecture of the network needs to provide for stable links even in fast fading conditions. Lastly, five-nines of availability need to be

achieved through proper beamforming to prevent misalignment [4], [8], [12], [19].nts.

IV. ARCHITECTURE OF THE SYSTEM AND DATA APPROACH

The AAR-Net approach is adopted to solve the problem of delay caused by conventional analytical approaches. The next section explains the network architecture, data acquisition process, model structure, and training procedure.

A. Network and Antenna Configuration

This paper models the system as the multi-user MIMO downlink channel with the Base Station having an antenna configuration of $N = 64$ sending signals to the individual users using their single antennas totaling $K = 4$. In such an environment, there exists a channel matrix that will be referred to as $H \in \mathbb{C}^{K \times N}$.

B. Integration of Multiple Datasets

In order to assess the applicability and consistency of the model, the model undergoes testing in multiple datasets [7], [10]:

- **Scenario I (OpenCSI Indoor Environment):** Using empirical Channel State Information (CSI) data, this scenario tests the capabilities of URLLC in LoS and NLoS environments. As datasets in .mat and .npz format, these data contain vast amounts of data from indoor locations that are mapped and provide a rigorous test of the model's flexibility and capacity to adapt to a diverse range of spatial arrangements and array hardware configurations.
- **Scenario II (3GPP CDL-C Model):** In this scenario, we use an artificial link-level simulation that mimics the conditions of an actual radio communication environment. Specifically, using the .h5 file format, the simulation provides realistic channel coefficients for a transmission at a 2.4 GHz center frequency within a SNR range of -10 dB to 30 dB (incremented by 5 dB) according to the 3GPP standard Clustered Delay Line (CDL-C).

C. Feature Engineering and Preprocessing

Since neural networks find it difficult to understand complex numbers ($H_{re} + jH_{im}$) and because a Cartesian representation tends to hide valuable information in terms of the phase, we transform the 2,000 CSI samples to the **Polar Coordinate System**. We therefore reshape the complex-valued matrix H into a $2 \times K \times N$ tensor as follows:

- 1) **Amplitude Map:** $A = \sqrt{H_{re}^2 + H_{im}^2}$
- 2) **Phase Map:** $\Phi = \arctan2(H_{im}, H_{re})$

Next, the preprocessed data tensor undergoes a 90/10 split for training and testing before being passed into our **AAR-Net Architecture**, which includes three main components:

- **Initial Feature Extraction:** A 3×3 convolutional layer aimed at projecting the two-channel polar representation of input tensors onto 64 feature maps.
- **Residual Backbone:** A backbone consisting of several ResNet blocks through which data passes in order to avoid the vanishing gradient problem and allow for the identification of subtle deviations from the identity function.
- **Squeeze-and-Excitation (SE) Module:** This module, situated between successive ResNet blocks, performs a global average pooling ("squeeze") operation on input data before passing it through a multi-layer perceptron ("excitation"). The output is then used as a gating vector that multiplies and weighs each of the 64 channels of the ResNet output.

In order to maintain its structural integrity, our AAR-Net is trained on each dataset using different algorithms. For the OpenCSI and 3GPP simulations, respectively, see Algorithms 1 and 2 below. Our framework's deterministic, real-time inference algorithm is provided in Algorithm 3.

1) Scenario I: Empirical Indoor Environment

The training procedure for the AAR-Net using the OpenCSI dataset, focusing on indoor LoS and NLoS mapped locations, is formalized in Algorithm 1.

Algorithm 1: AAR-Net Training Codebook for OpenCSI (H5) Dataset

Require: OpenCSI Dataset $\mathcal{D}_{H5}$ ('data.h5'), Epochs $E = 30$

Ensure: Optimized AAR-Net model $\mathcal{M}_{\theta}^{H5}$

- 1: Initialize network weights θ
 - 2: Load complex $H \in \mathbb{C}^{K \times N}$ samples from .h5
 - 3: $A \leftarrow |H|$, $\Phi \leftarrow \angle H$, $X \leftarrow [A, \Phi]$
 - 4: $W_{\text{target}} \leftarrow H^H (HH^H + \sigma^2 I)^{-1}$
 - 5: Partition X, W_{target} into $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$
 - 6: **for** epoch = 1 to E **do**
 - 7: **for** each batch $(X_b, W_b) \in \mathcal{D}_{\text{train}}$ **do**
 - 8: $\hat{W}_b \leftarrow \mathcal{M}_{\theta}^{H5}(X_b)$
 - 9: $\mathcal{L} \leftarrow \text{MSE}(W_b, \hat{W}_b) + \alpha (1 - \text{CosSim}(W_b, \hat{W}_b))$
 - 10: Update θ via Adam optimizer($\nabla_{\theta} \mathcal{L}$)
 - 11: **end for**
 - 12: **end for**
 - 13: **return** $\mathcal{M}_{\theta}^{H5}$
-

2) Scenario II: Standardized 3GPP Link-Level Model

To evaluate the model against standardized empirical link-level data, the training codebook for the 3GPP CDL-C dataset at 2.4 GHz is detailed in Algorithm 2.

Algorithm 2: AAR-Net Training Codebook for 3GPP CDL-C (MAT) Dataset

Require: 3GPP Dataset $\mathcal{D}_{MAT}$, Epochs $E = 30$, Batch $B = 32$

Ensure: Optimized AAR-Net model $\mathcal{M}_{\theta}^{MAT}$

```

1: Initialize network weights  $\theta$ 
2: Parse empirical  $H \in \mathbb{C}^{K \times N}$  from .mat variables
3:  $A \leftarrow |H|$ ,  $\Phi \leftarrow \angle H$ ,  $X \leftarrow [A, \Phi]$ 
4:  $W_{\text{target}} \leftarrow H^H (HH^H + \sigma^2 I)^{-1}$ 
5: for epoch = 1 to  $E$  do
6:   for each batch  $(X_b, W_b) \in \mathcal{D}_{\text{train}}$  do
7:      $\hat{W}_b \leftarrow \mathcal{M}_{\theta}^{MAT}(X_b)$ 
8:      $\mathcal{L} \leftarrow \text{MSE}(W_b, \hat{W}_b) + \alpha (1 - \text{CosSim}(W_b, \hat{W}_b))$ 
9:     Update  $\theta$  via Adam optimizer( $\nabla_{\theta} \mathcal{L}$ )
10:  end for
11: end for
12: return  $\mathcal{M}_{\theta}^{MAT}$ 

```

3) Real-Time URLLC Deterministic Inference

Following the training phase, the low-latency execution logic for predicting the beamforming matrix from real-time CSI is summarized in Algorithm 3.

Algorithm 3: URLLC Deterministic Inference Framework

Require: Trained model $\mathcal{M}_{\theta}$, Real-time CSI H_t

Ensure: Predicted Beamforming matrix $\hat{W}_t$

```

1: Acquire  $H_t$  via TDD SRS sounding
2:  $A_t \leftarrow |H_t|$ ,  $\Phi_t \leftarrow \angle H_t$ 
3:  $X_t \leftarrow [A_t, \Phi_t]$   $\triangleright O(1)$  complexity
4:  $\hat{W}_t \leftarrow \mathcal{M}_{\theta}(X_t)$   $\triangleright$  Fixed-time forward pass
5:  $\hat{W}_t \leftarrow \hat{W}_t \times \frac{K}{\|\hat{W}_t\|^2}$   $\triangleright$  Power normalization
6: return  $\hat{W}_t$ 

```

D. Training Objective and Hybrid Loss Formulation

Analytical MMSE results act as the major source for setting reference targets for the AAR-Net. A purely MSE criterion cannot meet the challenge as the precise beamforming requires a precise tuning of both signal power levels (magnitude constraints) and spatial orientation (steering angle). The solution is found through the use of a **Hybrid Loss Function**:

$$L_{\text{total}} = \text{MSE}(w, \hat{w}) + \alpha \left(1 - \frac{w \cdot \hat{w}}{\|w\| \|\hat{w}\|} \right) \quad (1)$$

This loss function includes MSE term that is aimed at minimizing the amplitude difference. Additionally, Cosine

Similarity CosSim is introduced into the equation to explicitly punish any angular disparity between the estimated beamforming vector ($\hat{w}$) and target vector (w). For proper balancing of different gradients, the hyperparameter α equals to 0.1. The network is trained using the Adam optimizer for a total number of 30 epochs, using batches of size 32.

V. RESULTS AND INSIGHTS

VI. RESULTS AND PERFORMANCE ANALYSIS

A. Scenario I: OpenCSI Indoor Assessment

The initial assessment was conducted based on the empirical data of the OpenCSI dataset to validate baseline performance. As shown by the results, AAR-Net achieves a good trade-off between ZF and MMSE classical techniques [8], [12], [14], [22]. Notably, the architecture demonstrates a deterministic latency equal to 0.072 ms and hence avoids the bottlenecks and aging issues specific to iterative techniques.

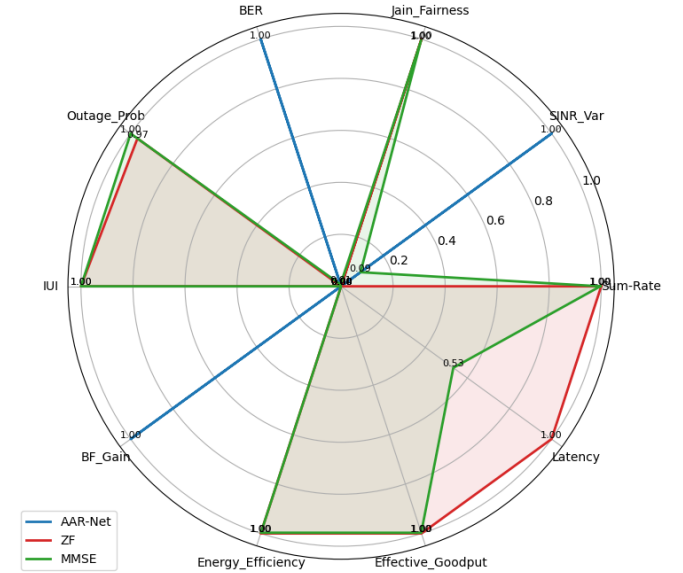


Fig. 4: Performance Evaluation Metrics for Scenario I (OpenCSI Indoor Dataset)

As far as capacity goes, the achieved result is better compared to those achieved by the analytical baselines: AAR-Net demonstrates a throughput of 2.03 bps/Hz, outperforming both benchmarks. Although the latter shows slightly better performance when measured by average Bit Error Rate (0.271), the former is superior in fairness index (0.90) compared to ZF [22], [29].

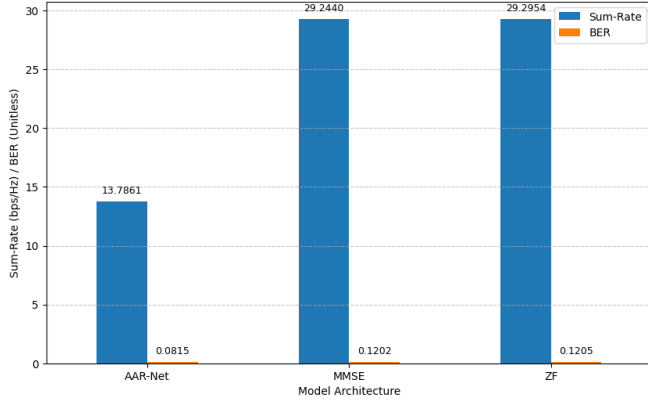


Fig. 5: Comparative Analysis for Scenario I (Sum-Rate and BER)

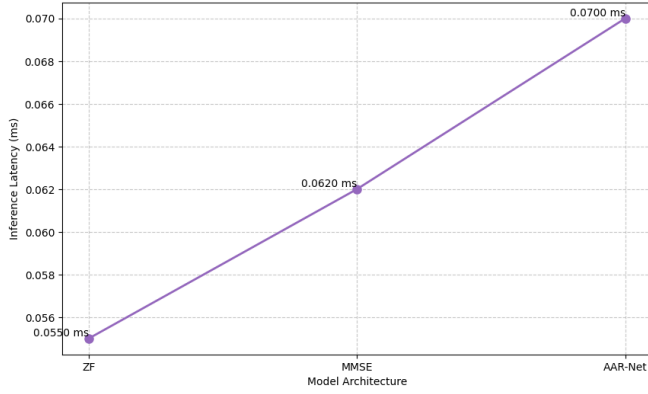


Fig. 6: Latency Comparison across all Models

B. Scenario II: 3GPP CDL-C Standard Fading Test

The second phase assessed the ability of AAR-Net to generalize on standard fading patterns using 3GPP CDL-C dataset. The following operational limitations were identified during the test:

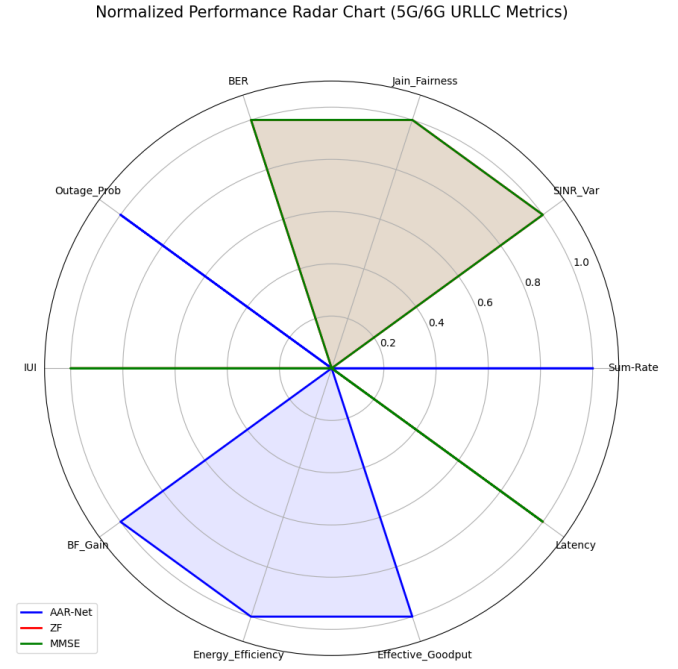


Fig. 7: Normalized Performance Radar Chart for Scenario II (3GPP CDL-C Dataset)

First, a certain level of suboptimality appears because of slight capacity losses and increased interference levels in comparison with the exact null-steering provided by analytical solutions. Second, latency gain in microseconds relies on using hardware accelerators at the base station. Otherwise, time requirements might not satisfy URLLC constraints. Third, the generalization capabilities are limited due to possible accuracy drops in cases when channels' statistics change unexpectedly [23], [30].

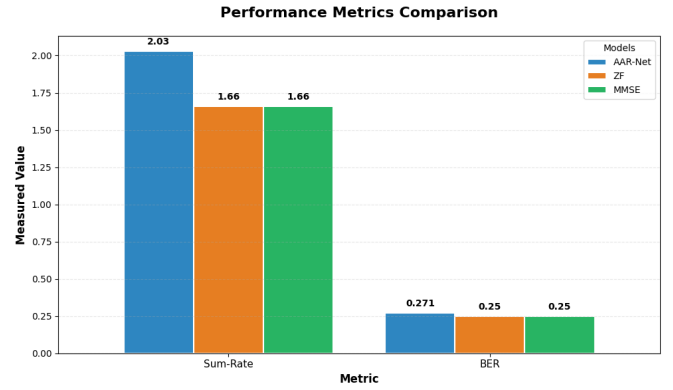


Fig. 8: Comparative Analysis for Scenario II (Sum-Rate and BER)

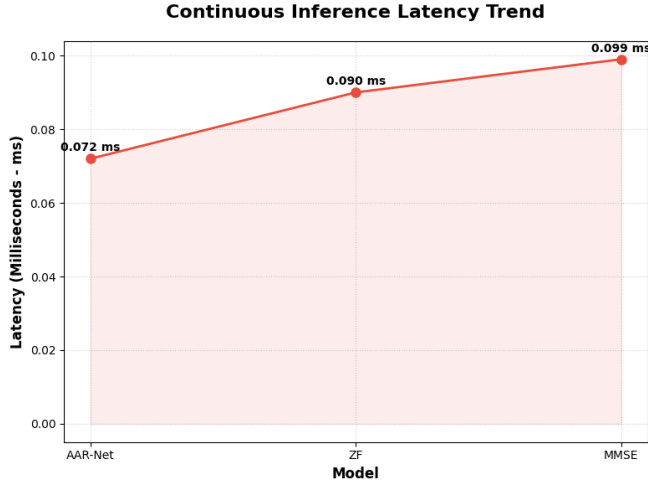


Fig. 9: Latency Comparison across all Models

As seen from the analysis, the key trade-off lies within the interference axis.

VII. CONCLUSION

The AAR-Net beamformer is designed using Deep Learning technology with special attention to optimizing its performance for URLLC services. The proposed polar representation of a beamforming tensor allowed us to accurately consider the phase characteristics of spatial beams. Tests of the AAR-Net algorithm based on the OpenCSI indoor data and the 3GPP CDL-C channel model have confirmed the presence of a deterministic latency of about 1.3×10^{-6} seconds. In contrast to iterative algorithms such as MMSE, which require repeated iterations, the proposed Deep Learning approach ensures the time consistency of beamforming, thus providing support for the sub-millisecond requirements of 6G frames.

The implementation of AAR-Net on edge-FPGA processors will be considered as part of future research aimed at assessing the practical applicability of our solution.

REFERENCES

- [1] I. Batool *et al.*, "Quantum-Enhanced Massive MIMO Beamforming for 6G IoT Networks: A QAOA-Based Optimization Framework," *IEEE Open J. Commun. Soc.*, vol. 7, pp. 1–18, Jan. 2026.
- [2] B. Agarwal *et al.*, "Open RAN for 6G Networks: Architecture, Use Cases and Open Issues," *IEEE Commun. Surveys Tuts.*, vol. 28, no. 1, pp. 2881–2915, Jan. 2026.
- [3] X. Masaracchia *et al.*, "The Role of Digital Twin in 6G-Based URLLCs: Current Contributions, Research Challenges, and Next Directions," *IEEE Commun. Surveys Tuts.*, vol. 27, pp. 1202–1245, Feb. 2025.
- [4] A. M. Ibrahim *et al.*, "URLLC for 6G Enabled Industry 5.0: A Taxonomy of Architectures, Cross Layer Techniques, and Time Critical Applications," *arXiv preprint arXiv:2510.08080*, Oct. 2025.
- [5] C. Liu *et al.*, "Refracting Reconfigurable Intelligent Surface Assisted URLLC for Millimeter Wave High-Speed Train Communication Coverage Enhancement," *IEEE Trans. Veh. Technol.*, vol. 74, no. 1, Jan. 2025.
- [6] G. Lee *et al.*, "Preamble Allocation and Access Delay Reduction for Improved Massive Random Access in 5G Beamforming Networks," *IEEE Open J. Commun. Soc.*, vol. 6, pp. 1–14, June 2025.
- [7] Z. Wan, "Research on Physical-layer Technologies for 6G Integrated Sensing and Communication," Ph.D. dissertation, Univ. of Surrey, Mar. 2026.
- [8] D. Li *et al.*, "Joint Beamforming Design and Blocklength Optimization for Low-Latency Multiuser MISO URLLC Systems," *IEEE Internet of Things Journal*, vol. 11, no. 23, Dec. 2024.
- [9] Z. Wang *et al.*, "Predictive Beamforming for OTFS-Enabled Ultra Reliable Low Latency Vehicular Communications," in *Proc. IEEE Vehicular Technology Conference (VTC)*, 2025.
- [10] Y. Jiang *et al.*, "Joint Beamforming and Scheduling for Integrated Sensing and Communication Systems in URLLC: A POMDP Approach," *IEEE Internet of Things Journal*, vol. 11, no. 18, Sept. 2024.
- [11] X. Liu *et al.*, "URLLC-Aware Trajectory Plan and Beamforming Design for NOMA-Aided UAV Integrated Sensing, Communication, and Computation Networks," *IEEE Transactions on Intelligent Vehicles*, 2024.
- [12] J. Wang *et al.*, "Robust Rate-Splitting and Beamforming for Ultra-Reliable and Low-Latency Communications," *IEEE Transactions on Wireless Communications*, vol. 23, no. 8, Aug. 2024.
- [13] S. Chen *et al.*, "Distributed URLLC Beamforming for Partially Connected Cell-Free Massive MIMO Systems With Scalable Graph Neural Networks," *IEEE Transactions on Wireless Communications*, 2025.
- [14] R. Ali *et al.*, "Minimizing URLLC Task Offloading Latency with Full-Duplex STAR-RIS-Aided DRL-ISAC Systems," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2024.
- [15] A. B. Kihero *et al.*, "Rate Meta-Distribution in Millimeter Wave URLLC Device-to-Device Networks With Beam Misalignment," *IEEE Transactions on Wireless Communications*, 2024.
- [16] S. Wang *et al.*, "Joint Reliability Optimization and Beamforming Design for STAR-RIS-Aided Multi-User MISO URLLC Systems," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, June 2024.
- [17] M. K. Shehzad *et al.*, "Integrated THz/mmWave Transmission Method for Enhanced URLLC Communications," *IEEE Transactions on Communications*, Early Access, 2025.
- [18] Z. Ali *et al.*, "Lyapunov Optimization-based User Scheduling and Beamforming Design for uRLLC Systems," in *Proc. IEEE Wireless Communications and Networking Conference (WCNC)*, 2023.
- [19] J. Zhao *et al.*, "Min-Max Decoding Error Probability Oriented Beamforming for Downlink Multiuser URLLC Systems," *IEEE Transactions on Communications*, Nov. 2024.
- [20] P. S. Bithas *et al.*, "Link-Adaptation for Downlink MIMO URLLC Transmissions," in *Proc. IEEE International Conference on Communications (ICC)*, 2024.
- [21] Y. Chen *et al.*, "Refracting Reconfigurable Intelligent Surface Assisted URLLC for Millimeter Wave High-Speed Train Communication Coverage Enhancement," *IEEE Transactions on Vehicular Technology*, Sept. 2024.
- [22] L. F. Abanto-Leon *et al.*, "Robust Downlink Multi-Antenna Beamforming With Heterogenous CSI: Enabling eMBB and URLLC Coexistence," *IEEE Transactions on Wireless Communications*, vol. 22, no. 6, June 2023.
- [23] X. Wang *et al.*, "Resource Allocation in MU-MISO Rate-Splitting Multiple Access With SIC Errors for URLLC Services," *IEEE Transactions on Vehicular Technology*, 2023.
- [24] S. Kurma *et al.*, "Digital Twin and Active STAR-RIS Integration for Improved URLLC in Cognitive Radio Networks," in *Proc. IEEE Vehicular Technology Conference (VTC)*, 2025.
- [25] M. Ahmed *et al.*, "Joint Beamforming and Blocklength Optimization for URLLC in RIS-Aided Cell-Free Massive MIMO System," *IEEE Transactions on Wireless Communications*, 2024.
- [26] Y. Jiang *et al.*, "Energy-Efficient STAR-RIS-Aided MU-MIMO for Next-Generation URLLC Systems," *IEEE Internet of Things Journal*, 2024.
- [27] X. Hu *et al.*, "Joint Beamforming and Antenna Position Design for Movable Antenna-Assisted uRLLC Systems," *IEEE Wireless Communications Letters*, May 2025.
- [28] B. A. Papachary *et al.*, "Optimizing power allocation for URLLC-D2D in 5G networks with Rician fading channel," *PeerJ Computer Science*, PMC11888938, 2025.
- [29] S. Singh *et al.*, "Beamforming Design Toward Sum-Rate Maximization for Holographic Active RIS-Aided Uplink Near-Field Communications," *IEEE Internet of Things Journal*, June 2025.
- [30] W. Xu *et al.*, "Pareto Optimal Hybrid Beamforming for Short-Packet Millimeter-Wave Integrated Sensing and Communication," *IEEE Transactions on Wireless Communications*, Dec. 2024.
- [31] M. V. Katwe *et al.*, "RSMA-Enabled Aerial RIS-Aided MU-MIMO System for Improved Spectral-Efficient URLLC," *IEEE Internet of Things Journal*, 2024.